CivicResolve : AI-Based Public Grievance Management System

A PROJECT REPORT

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October, 2025

BONAFIDE CERTIFICATE

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CHAPTER 1: INRODUCTION

The AI-Based Public Grievance Management System, CivicResolve, is a web application designed to help citizens submit public complaints and track their progress efficiently. Traditional grievance management systems may involve manual complaint classification, priority assignment, and limited transparency in decision-making. CivicResolve aims to address these challenges by providing a structured, systematic, and explainable approach to grievance handling.

The system uses Classical Symbolic Artificial Intelligence techniques, including Constraint Satisfaction Problem (CSP) and Forward Chaining. CSP validates the complaint details by checking predefined constraints and identifying inconsistencies. If the information is valid, the Forward Chaining rule-based inference engine applies predefined rules to recommend the appropriate department, priority level, and escalation requirement. A conflict-resolution mechanism handles competing priority recommendations, while a deterministic explanation engine presents the reasons behind the system's decisions in simple language.

CivicResolve supports three types of users: citizens, administrators, and department officers. Citizens can submit complaints, view updates, receive notifications, and monitor progress through a five-stage lifecycle: Submitted, Acknowledged, In Progress, Resolved, and Closed. Administrators can review complaints, assign departments, monitor progress, and verify resolutions, while department officers can manage assigned complaints and update their status.

The system focuses on explainability, transparency, consistency, and accountability in public grievance management. It uses predefined rules and structured complaint data rather than Machine Learning, Natural Language Processing, or Large Language Models. By combining symbolic AI reasoning with a web-based complaint tracking workflow, CivicResolve aims to support more organized and transparent grievance handling.

CHAPTER 2: LITERATURE SURVEY

2.1 Reimagining Digital Grievance Redressal Systems for Better Participatory Governance in India (2026)

This study examines citizen participation and communication in digital grievance management. It highlights the importance of transparency and citizen engagement but does not focus on rule-based AI for complaint prioritization and escalation. CivicResolve incorporates these concepts through complaint tracking, notifications, and communication history.

2.2 A Web-Based Public Grievance Redressal System with Image and Geolocation Support (2025)

This research presents a web-based grievance system supporting complaint registration, evidence uploads, location information, and status tracking. However, it does not describe the same combination of CSP validation and Forward Chaining used in CivicResolve. The proposed system extends these features with rule-based recommendations, conflict resolution, and deterministic explanations.

2.3 Explainable AI for Government: Does the Type of Explanation Matter to the Accuracy, Fairness, and Trustworthiness of an Algorithmic Decision as Perceived by Those Who Are Affected? (2024)

This study investigates how explanations influence people's perceptions of government algorithmic decisions. Although it emphasizes transparency and understandable decisions, it does not implement a complete grievance management system. CivicResolve uses a deterministic explanation engine to explain department, priority, and escalation recommendations based on the actual reasoning trace.

2.4 Explainable and Transparent Artificial Intelligence for Public Policymaking (2024)

This research discusses the importance of transparency and explainability in AI-supported public policymaking. Its focus is broader than the implementation of a public grievance workflow. CivicResolve applies these principles through predefined production rules and a reasoning trace that allows administrators to understand and review recommendations.

2.5 Does E-Service Quality of Online Grievance Redress Portals Lead to Satisfaction? An Outlook from the Perspectives of Indian Youth (2022)

This study examines the relationship between online grievance portal service quality and user satisfaction. It highlights the importance of a user-friendly digital public service but does not focus on AI-based complaint validation or priority determination. CivicResolve incorporates a citizen dashboard, progress indicators, notifications, and complaint history to improve the grievance tracking experience.

CHAPTER 3: PROPOSED SYSTEM

CivicResolve is a web-based public grievance management system designed to improve the way public complaints are validated, classified, assigned, and tracked. The proposed system combines classical symbolic Artificial Intelligence techniques with a structured workflow involving citizens, administrators, and department officers.

When a citizen submits a complaint through a structured form, the system first applies a Constraint Satisfaction Problem (CSP) to validate the selected values and identify inconsistent information. If the complaint passes validation, a Forward Chaining production rule engine uses predefined facts and IF–THEN rules to recommend the responsible department, priority level, and escalation requirement. If conflicting conclusions are produced, a conflict-resolution mechanism applies the predefined priority order. A deterministic explanation engine then presents the reasons behind the recommendations in understandable language.

The system does not use Machine Learning, Natural Language Processing, or Large Language Models. Instead, it uses predefined rules and structured inputs to provide consistent, traceable, and explainable recommendations.

After the AI assessment, the administrator reviews and acknowledges the complaint, assigns it to the appropriate department, and monitors its progress. The department officer processes the complaint, provides updates, and marks it as resolved. The administrator verifies the resolution before closing the complaint. Citizens can monitor the entire process through a progress indicator, notifications, status updates, and a complaint timeline.

The proposed system aims to improve transparency, reduce inconsistent manual classification, support timely escalation, and maintain accountability throughout the grievance redressal process.



Fig 3.1 Architecture Diagram

CHAPTER 4: MODULE DESCRIPTION

CivicResolve is divided into the following modules to make the application easier to develop, maintain, and test.

4.1 Module 1 – User Authentication and Role Management

This module manages secure access to the application. It supports three roles: Citizen, Administrator, and Department Officer. Each user is provided access to the features required for their responsibilities.

Citizens can register, log in, submit complaints, and track their own grievances. Administrators can review all complaints, manage assignments, maintain rules, and monitor the overall workflow. Department officers can view and update complaints assigned to their respective departments.

Role-based access control prevents unauthorized users from performing restricted actions. Passwords should be stored securely using a suitable password-hashing algorithm.

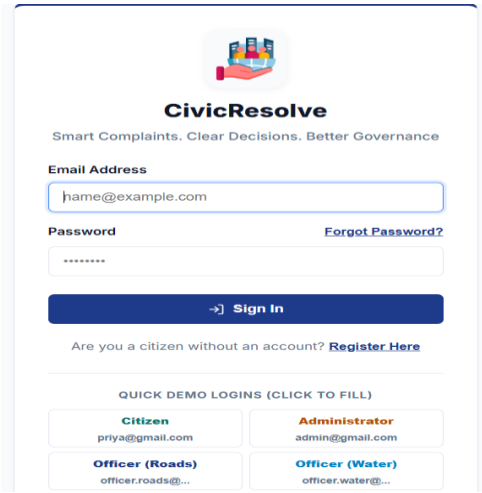


Fig 4.1 Login Page

4.2 Module 2 – Complaint Registration and CSP Validation

This module allows citizens to submit complaints through a structured form containing fields such as category, issue type, severity, immediate danger, safety risk, frequency, and population impact.

The CSP validation component checks whether the selected values belong to their permitted domains and satisfy predefined constraints.

For example, if a citizen reports immediate danger but selects a low severity level, the system can identify the combination as inconsistent if that constraint is defined in the knowledge base. It then reports the conflict and requests administrative review rather than proceeding with an unsupported recommendation.

The module ensures that only complaints satisfying the required validation conditions proceed to normal AI reasoning. Valid complaint information is converted into facts for the inference engine.

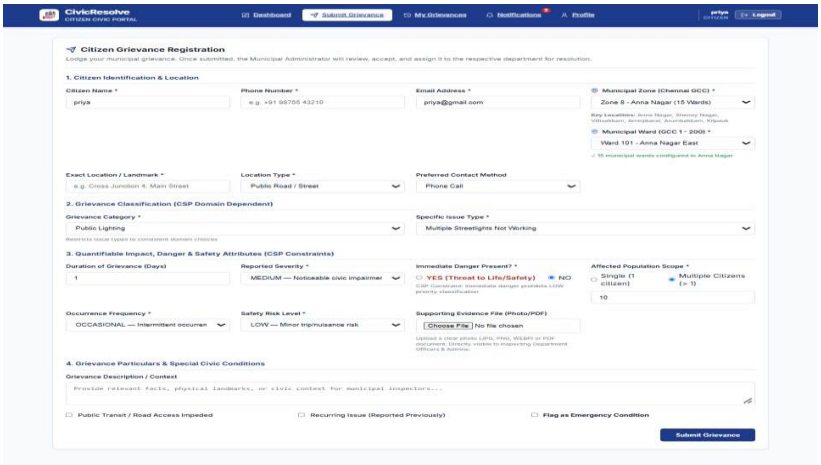


Fig 4.2 Complaint Module

4.3 Module 3 – Knowledge Base and Forward Chaining Inference Engine

This module contains the knowledge base and the main AI reasoning mechanism. The knowledge base stores the predefined facts, rules, and domain information required to assess complaints.

The Forward Chaining engine begins with verified facts and matches them against applicable IF–THEN production rules. When a rule's conditions are satisfied, its conclusion is added to working memory. The engine continues applying applicable rules until no new conclusions can be derived.

For example:

- Rule condition: The complaint category is Water, and the issue type is Pipeline Leakage.
- Rule conclusion: Recommend the Water Supply Department.

Other rules can consider severity, immediate danger, frequency, safety risk, and population impact to determine priority and escalation requirements.

The final output contains the recommended department, priority level, and escalation status. These outputs are recommendations based on predefined rules, not predictions generated by a machine-learning model.

4.4 Module 4 – Conflict Resolution and Deterministic Explanation

This module handles competing conclusions and explains the system's recommendations.

When multiple applicable rules produce different priority levels, the conflict-resolution mechanism follows the predefined precedence order:

CRITICAL > HIGH > MEDIUM > LOW

This ensures that the system resolves priority conflicts consistently. Validation conflicts, such as contradictory complaint inputs, are handled separately and may require administrative review.

The deterministic explanation engine uses the facts, rules fired, and conclusions derived by the inference

engine to produce a readable explanation. It explains why a department was recommended, why a priority was selected, and whether escalation is required.

The explanation is generated from predefined templates and the actual reasoning trace. It does not require an LLM or NLP system.

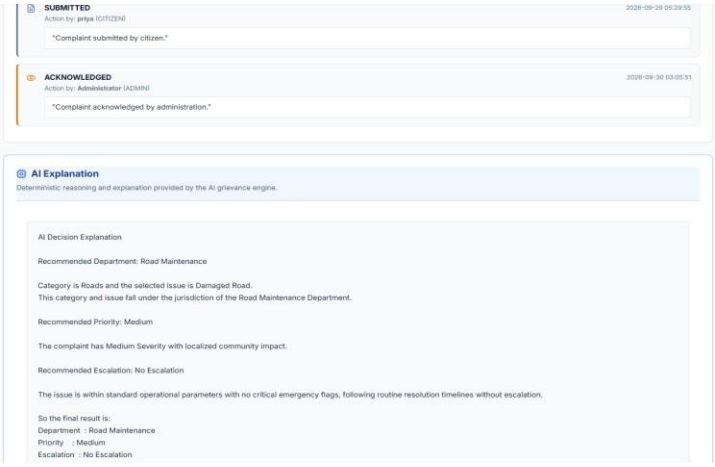


Fig 4.3 AI Explanation Page

4.5 Module 5 – Complaint Assignment and Department Officer Management

This module manages the operational processing of complaints after AI assessment.

The administrator reviews the AI recommendation, acknowledges the complaint, and assigns it to the responsible department. If the recommendation is unsuitable, the administrator can override the department and record the reason for the change.

Department officers can view their assigned complaints, start processing them, add progress updates, communicate relevant information, and mark complaints as resolved.

The administrator verifies the reported resolution before closing the complaint. This arrangement ensures that AI recommendations support the workflow without replacing human responsibility.

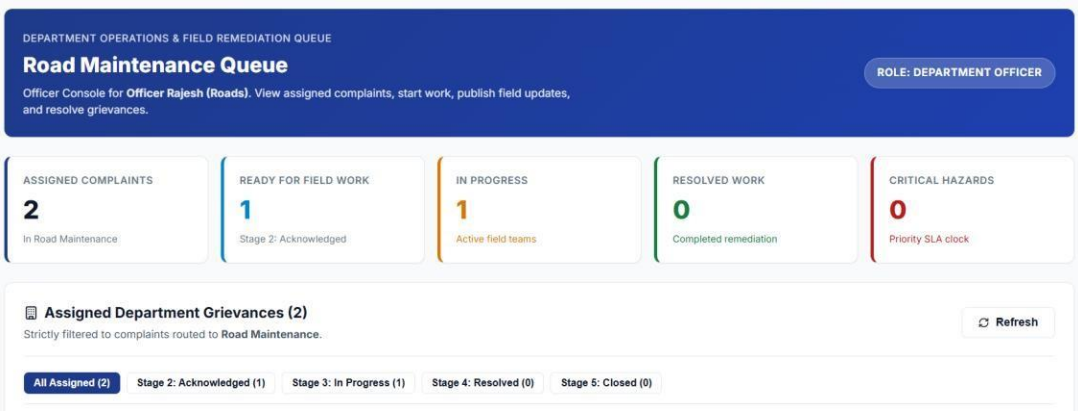


Fig 4.4 Department Module

4.6 Module 6 – Complaint Tracking, Notifications and Dashboard

This module provides status visibility to citizens, administrators, and department officers.

The complaint lifecycle consists of five stages:

Stage	Progress	Description
Submitted	20%	Complaint registered
Acknowledged	40%	Administrator acknowledges the complaint
In Progress	60%	Department officer begins processing
Resolved	80%	Officer reports the resolution
Closed	100%	Administrator verifies and closes the complaint

The percentages represent the defined workflow stages, not measured completion of the actual repair or service work.

The citizen dashboard displays complaint counts, current status, progress, notifications, and the complaint timeline. The administrator dashboard provides complaint queues, departmental workload information, priority-based filtering, and access to audit records.

The system enforces valid sequential transitions and rejects unauthorized or invalid status changes. Notifications and timeline entries help maintain a record of important complaint updates.

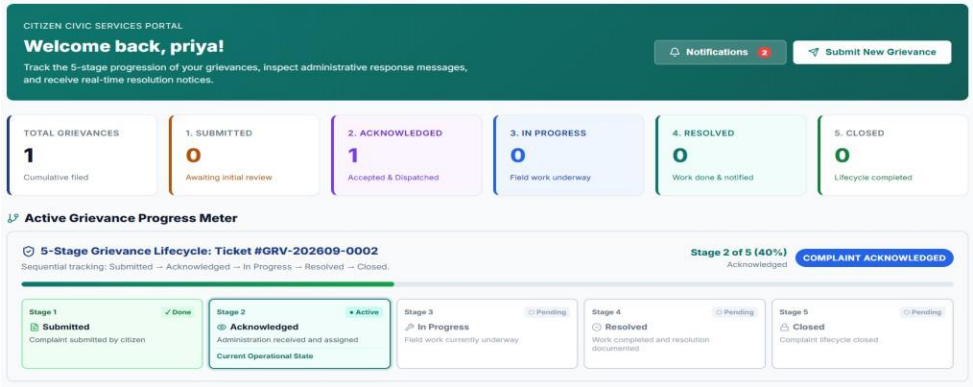


Fig 4.5 Dashboard

CHAPTER 5: IMPLEMENTATION AND RESULTS

The implementation of CivicResolve consists of a web-based user interface, a backend application, a database, and a symbolic AI reasoning subsystem. The frontend handles user interaction, the backend manages complaint processing and business logic, and the AI subsystem validates complaint information and generates explainable recommendations.

5.1 Experimental Setup

The experimental setup defines the software environment and testing procedures used to implement and evaluate the proposed system.

Software requirements

Component	Proposed Technology
Frontend	React.js, JavaScript, HTML and CSS
Backend	Python with FastAPI
Database	SQLite 3
AI Implementation	Python
Validation Algorithm	Constraint Satisfaction Problem (CSP)
Reasoning Algorithm	Forward Chaining production rule engine
Development Environment	Visual Studio Code
Testing Tools	Pytest and FastAPI TestClient

Experimental procedure

The system is evaluated through the following steps:

1. Submit complaints containing valid and invalid combinations of structured field values.
2. Verify that CSP identifies domain violations and predefined constraint conflicts.
3. Submit valid complaints and check whether Forward Chaining derives the expected department, priority, and escalation level.
4. Test whether the conflict-resolution mechanism consistently selects the highest applicable priority.
5. Verify that explanations correspond to the facts and rules used during reasoning.
6. Test complaint assignment, role permissions, status transitions, and administrator overrides.
7. Verify that complaint updates appear in the citizen dashboard and that notifications and timeline records are maintained.
8. Test invalid requests, missing information, and unauthorized actions to ensure that the system handles them appropriately.

5.2 Results

The results section should describe the behaviour observed during testing. The following table provides a test plan and the expected behaviour of the proposed system. These are expected outcomes, not claimed experimental results.

Test Case	Input / Action	Expected Result
CSP Validation	All required fields contain valid, consistent values	Complaint proceeds to AI reasoning
Constraint Conflict	Immediate danger is Yes, but severity is Low, where a corresponding constraint is defined	Conflict is reported and automatic recommendation is paused
Department Recommendation	A valid Water complaint matches a Water Supply routing rule	Water Supply is recommended
Priority Determination	Verified facts match a predefined high-priority rule	The priority derived by the rule is returned
Priority Conflict	Different applicable rules derive HIGH and CRITICAL	CRITICAL is selected according to the precedence order
Explanation Generation	Rules derive department, priority, and escalation	Explanation reflects the actual reasoning trace
Role-Based Access	Citizen attempts to access an administrative function	Access is denied
Invalid Status Transition	A user attempts to skip a required lifecycle stage	The transition is rejected
Department Assignment	Administrator assigns a complaint to an officer's department	Assigned complaint appears in the officer's work queue
Complaint Closure	Administrator verifies a reported resolution	Complaint moves to CLOSED and progress reaches 100%

Expected outcome

The proposed system is expected to provide consistent complaint validation, reproducible rule-based recommendations, and understandable explanations. The structured workflow is intended to improve complaint traceability and make responsibilities clearer for citizens, administrators, and department officers.

After implementation, the actual results should be recorded using the number of test cases passed, the number of failures, and any defects identified. If the project is evaluated using quantitative measures, report the measured values and the testing procedure rather than claiming accuracy or efficiency improvements without evidence.

CHAPTER 6: CONCLUSION AND FUTURE WORKS

CivicResolve is a proposed AI-based public grievance management system that combines classical symbolic AI with a structured web application. The system uses Constraint Satisfaction Problem validation to identify invalid or inconsistent complaint information and Forward Chaining to derive department, priority, and escalation recommendations from predefined production rules. A conflict-resolution mechanism handles competing priority conclusions, while a deterministic explanation engine presents the reasoning in understandable language.

The application also supports citizen complaint registration, administrator review, department-officer processing, progress tracking, notifications, and resolution verification. Its five-stage lifecycle provides a structured approach to complaint management, while the audit trail supports accountability and traceability. Unlike systems that depend on Machine Learning or Large Language Models, CivicResolve relies on predefined facts, constraints, and rules. This makes its reasoning process explicit and reproducible. The proposed design aims to support consistent and transparent grievance handling while keeping final operational decisions under human supervision. Its practical effectiveness will be established through implementation and systematic testing.

Future Work

The following enhancements can be considered in future versions:

1. Expanded knowledge base: Add and maintain more verified rules for different public-service categories and departments.
2. Rule maintenance interface: Allow authorized administrators to manage routing and priority rules with validation and audit logging.
3. Geolocation support: Capture complaint locations to help identify the responsible local authority.
4. Improved accessibility: Add multilingual interfaces and accessibility features for a wider range of citizens.
5. Advanced analytics: Generate reports on complaint resolution time, recurring issues, pending complaints, and departmental workload using actual database records.
6. Security enhancements: Strengthen authentication, authorization, data protection, and audit mechanisms.
7. Evaluation with real cases: Evaluate rule coverage, conflict detection, recommendation consistency, usability, and processing time using a carefully prepared test dataset.
8. Optional future text processing: If free-text complaint classification is later required, evaluate NLP as a separate extension without making it a dependency of the current symbolic reasoning architecture.

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